

Combining Deep Learning With RL

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Navigating the Grid World and Beyond



Analyx

	a_1	a_2	a_3	a_4
s_1	$Q(s_1, a_1)$	$Q(s_1, a_2)$	$Q(s_1, a_3)$	$Q(s_1, a_4)$
s_2	$Q(s_2, a_1)$	$Q(s_2, a_2)$	$Q(s_2, a_3)$	$Q(s_2, a_4)$
s_3	$Q(s_3, a_1)$	$Q(s_3, a_2)$	$Q(s_3, a_3)$	$Q(s_3, a_4)$
s_4	$Q(s_4, a_1)$	$Q(s_4, a_2)$	$Q(s_4, a_3)$	$Q(s_4, a_4)$

-1_{s_1}	-1_{s_2}	-1_{s_3}
-1_{s_4}	-1_{s_5}	-1_{s_6}
-1_{s_7}	-10_{s_8}	$+10_{s_9}$



What happens when the state-space is too large!



Q-table will be too big to fit into Analyx's tiny brain



Use Q-table with deep learning

Thank You
